The Five-Day Workweek and Spatial Labor Supply: Evidence from South Korea

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Abstract

This paper studies whether working-time regulation affects residential location choices by changing the effective cost of commuting. We exploit the staggered statutory introduction of the Five-Day Workweek in South Korea, which reduced the standard workweek from six to five days, and assign workers to legal eligibility cohorts based on the size and sector of their entry firm. Using individual-level panel data, we focus on short-tenure workers employed in Seoul, whose job–residence arrangements are relatively less settled. We find that statutory eligibility increases the probability of residing in Gyeonggi by about 5 percentage points, with no systematic pre-eligibility pattern in the event study. The response is concentrated on the Seoul–Gyeonggi residential margin and is larger among workers who were already residing outside Seoul when entering their current job. These findings show that working-time regulation can induce spatial adjustment within a metropolitan labor market by altering commuting incentives.

Keywords: Residential Location; Spatial Labor Supply; Commuting; Working-Time Regulation.

JEL Classification: J22, R23, R41.

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1 Introduction

Over the past decades, a growing literature has emphasized the importance of spatial adjustment in labor markets. Workers respond to economic incentives not only by adjusting hours or labor force participation, but also by reallocating across locations through commuting and residential choices. While the advantages of larger cities attract workers and firms (*e.g.*, De La Roca and Puga (2017); Henderson (2003); Glaeser et al. (2001)), spatial equilibrium is shaped simultaneously by dispersion forces such as congestion, housing costs, and commuting disutility. As a result, labor supply decisions are inherently spatial, and understanding how institutional changes affect these margins is central to modern urban and labor economics.

A key component of this spatial trade-off is commuting. Higher urban wages and productivity can sustain employment in dense employment centers, while housing costs create incentives to reside farther away. The attractiveness of peripheral residence therefore depends in part on the cost of maintaining access to urban jobs. Changes in commuting costs can consequently affect residential location even without direct changes in wages or housing prices.

Consistent with this view, existing empirical studies show that commuting costs materially affect labor supply decisions. Longer commuting distances and times have been shown to influence both the intensive margin, such as working hours, and the extensive margin, such as labor force participation (*e.g.*, Agarwal et al. (2018); Black et al. (2014); i Puigarnau and van Ommeren (2010)). More recently, the literature has emphasized that commuting frictions shape the responsiveness of local labor markets to economic shocks. Monte et al. (2018), for example, show that commuting links shape how local employment responds to labor-demand shocks, highlighting the broader importance of commuting frictions for spatial labor-market adjustment.

Despite these advances, there is still limited evidence on how institutional changes that alter commuting intensity, rather than wages or employment directly, affect residential location choices within local labor markets. In particular, less is known about whether working-time institutions that can alter the burden of commuting affect where workers live relative to their workplaces.

This paper studies how the introduction of the Five-Day Workweek in South Korea affected residential location choices within the Seoul metropolitan labor market. The reform reduced statutory weekly hours from 44 to 40 and institutionalized a five-day workweek for covered workers, thereby plausibly reducing the commuting burden associated with regular work schedules This makes the policy a useful setting for studying whether working-time regulation affects residential location through commuting-related incentives. To the best of our knowledge, no existing study examines the reform through a spatial lens or asks whether working-time regulation affects workers' residential location relative to their workplaces.

Building on Moretti (2011), we use a simple residential location framework in which commuting costs depend on the number of commuting days. A reduction in commuting days lowers the relative cost of residing farther from the employment center while maintaining access to

the same labor market. In the Seoul metropolitan area, this mechanism predicts an increase in Seoul workplace–Gyeonggi residence matches.

A key empirical challenge is that statutory coverage of the Five-Day Workweek was phased in by firm size and sector, creating systematic differences in eligibility timing across workers. Our empirical strategy exploits this legal rollout by assigning each worker an eligibility year based on the size and sector of the worker's entry firm. This entry-based assignment fixes eligibility timing at the worker level and anchors event time to the statutory schedule, rather than allowing exposure to vary with subsequent job changes.

Empirically, we use individual-level panel data to study short-tenure workers employed in Seoul—workers observed within their first three years at the current job (tenure 0–2), whose residential choices are plausibly less constrained by legacy job–housing matches. Using an event-time specification around statutory eligibility, we examine how the probability of residing in Gyeonggi while working in Seoul evolves around legal coverage. We also examine heterogeneity by marital status, entry wage, and residence at current-job entry, and decompose the residential response across Seoul, Gyeonggi, and other regions.

We find that short-tenure workers employed in Seoul become about 5–6 percentage points more likely to reside in Gyeonggi after statutory eligibility begins. The event-study estimates show little systematic evidence of differential pre-eligibility trends, while the post-eligibility coefficients are positive and jointly significant. The response is concentrated on the Seoul–Gyeonggi residential margin. Heterogeneity estimates suggest non-uniform adjustment across workers, with the clearest differences appearing by residence at current-job entry.

This paper contributes to the literature in two ways. First, it adds a spatial dimension to the literature on working-time regulation. Existing studies have primarily examined the effects of shorter workweeks on labor supply, employment, productivity, and worker well-being. This paper shows that statutory eligibility for working-time reduction can also affect residential location choices within a metropolitan labor market, highlighting a spatial margin through which labor-market institutions may operate. Second, the paper provides evidence on residential adjustment near the beginning of a job match, when job–housing arrangements are plausibly less settled. Focusing on short-tenure workers employed in Seoul, the analysis shows that the response to statutory eligibility is concentrated on the Seoul–Gyeonggi residential margin. The heterogeneity analysis further suggests that adjustment differs across workers' initial residential positions, while other subgroup differences are less precisely estimated.

Beyond its academic contribution, the paper also speaks to ongoing policy debates on further reductions in working time. While shorter-workweek reforms are typically evaluated through their effects on employment, productivity, and worker well-being, the results suggest that they may also have spatial consequences through residential location choices. This spatial dimension is relevant for discussions of additional reforms, such as proposals for a four-and-a-half-day or four-day workweek, and highlights the importance of considering spatial adjustment when evaluating the broader effects of labor-market institutions.

2 Institutional Background

Korea began implementing a five-day workweek in the mid-2000s as part of a broader effort to reduce excessive working hours and improve work–life balance. The policy was motivated by concerns about long working hours and was also discussed in connection with potential employment and productivity effects.

In the early 2000s, the Economic and Social Development Commission formed a Special Committee on Workweek Reduction to evaluate a reduction in standard hours. Following negotiations among labor unions, employer associations, and the government, revisions to the Labor Standards Act were proposed and subsequently passed by the National Assembly in 2003. The reform reduced standard weekly hours to 40 while maintaining the statutory limit on overtime work, and it was accompanied by provisions intended to mitigate nominal wage reductions. (See Ahn (2022))

Table 1 summarizes the statutory rollout schedule. Coverage was determined primarily by establishment size and sector, with large establishments and the public/financial sectors covered first and progressively smaller establishments incorporated over time. This phased implementation generated substantial variation in statutory eligibility timing across firms, which we exploit in the empirical analysis.

Table 1: Staggered Rollout of the Five-Day Workweek in Korea

Implementation year	Coverage	Notes
2004	$\geq$ 1,000 employees; finance and public sector	First-stage adoption
2005	$\geq$ 300 employees	Large private establishments
2006	$\geq$ 100 employees	Medium-sized establishments
2007	$\geq$ 50 employees	Expansion to smaller firms
2008	$\geq$ 20 employees	Broad coverage begins
2011	$\geq$ 5 employees	Near-universal application

The empirical analysis exploits statutory eligibility timing implied by this rollout schedule. Specifically, each worker is assigned an eligibility year based on the size and sector of the worker's entry firm under the legal schedule. This entry-based assignment fixes eligibility timing at the worker level and provides the basis for the event-time analysis below.

For the purposes of this paper, the reform is informative not only because it reduced statutory weekly hours, but also because the transition to a five-day workweek plausibly reduced the regular commuting burden associated with covered jobs. This feature motivates interpreting the reform as a change in commuting-related incentives in the theoretical and empirical analysis that follows.

3 Theoretical Framework

This section develops a simple spatial-equilibrium model in the spirit of Moretti (2011). I augment the framework with a commuting-days wedge to capture the policy-induced reduction in commuting intensity associated with the legal transition to the five-day workweek. There are two residential locations, $r \in \{S, G\}$ (Seoul and Gyeonggi), and one job location, $j = S$ (workplace in Seoul), reflecting the empirical focus on short-tenure workers employed in the Seoul labor market. [1]

A policy reduces the number of commuting days required for Seoul jobs from $N_0 = 6$ to $N_1 = 5$. I interpret this as a reduction in effective commuting costs.

As in Moretti (2011), I assume that each city is a competitive economy that produces a single output good y, with price normalized to one. Also, workers and firms are mobile and locate where utility and profits are maximized.

3.1 Residential Location Choice

First, suppose the indirect utility of worker i residing in city r and working in j is

$$U_{irj} = w_j - R_r + A_r - \tau_{rj}N + \varepsilon_{irj} \tag{1}$$

where w_j is the wage in workplace city j, R_r is the housing cost in city r, and A_r is the non-commuting amenity in city r. Also, the commuting cost is incorporated as $\tau_{rj}N$ where τ_{rj} is the per-day commuting cost from r to j and N is the number of commuting days.[2] The policy effect is captured as a reduction in N from N_0 to N_1.

The random term ε_{irj} denotes an idiosyncratic preference shock. For simplicity, I assume that ε_{irj} follows an i.i.d. Uniform distribution on $[-\bar{\varepsilon}, \bar{\varepsilon}]$.[3] Let the relative deterministic utility of living in Gyeonggi versus Seoul be

$$\Delta V = (A_G - A_S) - (R_G - R_S) - (\tau_{GS} - \tau_{SS})N \tag{2}$$

Define $s \equiv \Pr(\text{live in G})$ among workers employed in Seoul.

3.2 Labor Market

Let the production function for firms in city j be Cobb-Douglas

$$\ln y_j = X_j + hN_j + (1 - h)K_j \tag{3}$$

[1] The model can be extended to multiple job locations; conditioning on $j = S$ is sufficient to map the model to empirical design.

[2] By the Korean context, it is natural to assume that $\tau_{GS} > \tau_{SS}$.

[3] Moretti (2011) also considers a uniform specification, although one could alternatively assume a type-I extreme-value distribution, as in the standard logit model, to obtain closed-form choice probabilities.

where X_j is a city-specific productivity shifter, N_j is log labor input, and K_j is log capital. Since I only consider the workplace in Seoul, I drop the j subscript hereafter.

By the zero-profit condition, the wage in city j is given by

$$w = X - (1-h)P + (1-h)K + \ln h \tag{4}$$

which can be interpreted as the inverse labor-demand equation.

3.3 Housing Market

As in Moretti (2011), the housing supply in city r is assumed to be given by

$$R_r = \bar{R}_r + \kappa_r P_r, \qquad \kappa_r > 0, \qquad r \in \{Gyeonggi, Seoul\} \tag{5}$$

where P_r is the residential population in city r and κ_r represents the inverse elasticity of housing supply. Also, $\bar{R}_r$ is the exogenous fixed housing cost in city r.

3.4 Equilibrium

Assume that X, K, and A_r are fixed and exogenously given. The equilibrium is characterized by (L, s, w, R_S, R_G) such that: (i) labor market clears: w satisfies (4), (ii) rents satisfy (5) given N_r, (iii) s is consistent with the residential-choice probability implied by (1).[4]

Recall that $\varepsilon \sim U[-\bar{\varepsilon}, \bar{\varepsilon}]$. Then, from (1) and (2), a worker chooses to live in Gyeonggi if and only if $\Delta V + \eta_i \leq 0$ where $\eta_i = \varepsilon_{iGS} - \varepsilon_{iSS} \sim U[-2\bar{\varepsilon}, 2\bar{\varepsilon}]$. Hence, in the interior region (that is, away from corner solutions), the probability of living in Gyeonggi is linear.

$$s = \Pr(\eta_i \geq -\Delta V) = \frac{1}{2} + \frac{\Delta V}{4\bar{\varepsilon}}, \qquad \text{for } |\Delta V| < 2\bar{\varepsilon}. \tag{6}$$

This delivers a tractable local residential-supply response, as in Moretti (2011).

Using (5), (2) and (6), the equilibrium satisfies

$$w = X - (1-h)N + (1-h)K + \ln h \tag{7}$$

$$R_G - R_S = (\bar{R}_G - \bar{R}_S) + \kappa_G(sL) - \kappa_S((1-s)L) \tag{8}$$

$$s = \frac{1}{2} + \frac{1}{4\bar{\varepsilon}}\left[(A_G - A_S) - (R_G - R_S) - (\tau_{GS} - \tau_{SS})N\right] \tag{9}$$

[4] Since I focus on residential location choice while taking wages, employment, and housing prices as given, this framework is partial equilibrium.

3.5 Comparative Statics

Recall that X, K, and A_r are fixed.

Proposition 1 (Residential-share response under Uniform shocks)**.** *In the interior region where (6) holds, a reduction in commuting days ($N \downarrow$) increases the equilibrium share of Seoul workers residing in Gyeonggi:*

$$\frac{\partial s}{\partial N} < 0.$$

Moreover, the magnitude of the response is larger when taste dispersion is smaller (smaller $\bar{\varepsilon}$) and when the commuting-cost gap is larger (larger $\tau_G - \tau_S$).

From (9), holding rents fixed, $\partial s/\partial N = -(\tau_{GS} - \tau_{SS})/(4\bar{\varepsilon}) < 0$. Allowing rents to adjust through (8) dampens the response: as s increases, R_G rises relative to R_S (especially if κ_G is large), offsetting some of the commuting-cost gain. However, the sign remains negative because the direct effect of N on ΔV is negative for G relative to S when $\tau_{GS} > \tau_{SS}$.

This also implies that the increase in s induced by $N \downarrow$ is smaller when housing supply in Gyeonggi is more inelastic (larger κ_G), as rents in G (Gyeonggi) rise more sharply with inflows. The policy acts like an improvement in the attractiveness of living farther from the job center. In equilibrium, part of the benefit is capitalized into higher rents in the destination location (Gyeonggi) when housing supply is inelastic, paralleling the incidence logic in Moretti (2011).

4 Data

4.1 Data Source and Sample Selection

My main dataset is the Korea Labor and Income Panel Study (KLIPS), a nationally representative annual household panel survey conducted by the Korea Labor Institute since 1998. KLIPS provides detailed information on individual demographics, employment, earnings, workplace location, residential location, and labor-market transitions, along with person-level sampling weights.

The Five-Day Workweek reform was phased in between 2004 and 2011 according to establishment size and sector. To capture both pre-reform patterns and post-reform adjustment, I use KLIPS waves from 2001 to 2015. This window provides several years of pre-coverage observations and allows medium-run residential adjustment to unfold.

I restrict attention to wage workers and exclude the self-employed, whose work schedules, labor-market attachment, and exposure to statutory working-time rules differ systematically from those of employees. I further restrict the sample to workers whose workplace is located in Seoul. This focuses attention on a common labor market and sharpens the residential margin of interest, namely the Seoul–Gyeonggi margin, where commuting links are particularly strong. At the same time, this restriction should not be interpreted as eliminating Seoul–Gyeonggi-specific housing-market or transportation shocks. The resulting unbalanced panel contains 3,020 workers and 14,250 person-year observations.

All summary statistics and regressions use KLIPS person weights. The main analysis further restricts the Seoul-workplace sample to short-tenure workers, defined below.

4.2 Short-Tenure Workers

A central challenge in studying spatial adjustment is the substantial inertia in residential and commuting choices among incumbent workers. Housing contracts, family constraints, and job-specific attachment can all limit short-run relocation, even when commuting conditions change.

To focus on a margin where residence–workplace arrangements are plausibly less settled, I restrict attention to short-tenure workers in the Seoul workplace sample. I define *short-tenure workers* as wage workers employed in Seoul whose tenure in the current job is between 0 and 2 years, where tenure is measured as survey year minus job start year. This definition includes both first-time entrants and workers who recently changed jobs.

The purpose of this restriction is not to isolate first-time labor-market entry per se, but rather to study workers near the beginning of a new job match, when residential adjustment is more likely to remain feasible. In the Seoul labor market, where housing costs are typically higher than in Gyeonggi[5], a reduction in weekly commuting frequency can make peripheral residence relatively more attractive during the early phase of a match.

The resulting short-tenure Seoul-workplace sample contains 6,178 person-year observations.

4.3 Descriptive Statistics

Table 2 reports weighted summary statistics for wage workers employed in Seoul. Columns (1)–(2) summarize the full Seoul-workplace sample, while Columns (3)–(4) restrict to short-tenure workers with tenure 0–2 years in their current job. Panel A reports individual characteristics, Panel B job characteristics, and Panel C housing characteristics.

Short-tenure workers are younger, less likely to be married, and earn lower wages than the full sample, consistent with the fact that they are earlier in their career trajectories. Their average housing contract type differs, consistent with greater reliance on rental arrangements, although the categorical coding warrants cautious interpretation. Taken together, these patterns suggest that short-tenure workers are positioned on a less-settled residence–workplace margin and may therefore be more responsive to changes in commuting conditions.

[5]This spatial trade-off is also consistent with official regional housing statistics from the Korea Real Estate Board, *National Housing Price Trends Survey, Monthly Average Housing Sale Price (All Housing Types)*, which show that average housing sale prices were substantially higher in Seoul (454,531,000 KRW) than in Gyeonggi (261,948,000 KRW) during 2012–2015, i.e., in the later part of our sample period.

Table 2: Summary Statistics

	Full sample		Short-tenure workers	
	Mean	SD	Mean	SD
Panel A. Individual Characteristics				
Age (years)	39.332	12.132	35.352	12.132
Education Level (categorical)	5.972	1.345	5.973	1.275
Male (=1)	0.566	0.496	0.477	0.500
Married (=1)	0.558	0.497	0.387	0.487
Log wage	5.108	0.678	4.865	0.633
Panel B. Job Characteristics				
Weekly average working days	4.930	1.340	4.913	1.406
Weekly average working hours	42.575	18.453	41.793	20.183
Panel C. Housing Characteristics				
Housing tenure type	1.513	0.758	1.564	0.788
Housing structure type	2.336	1.142	2.345	1.197
Floor space (pyeong)	30.622	16.349	29.634	16.133
Housing price (10,000 KRW)	28990.075	25133.807	27286.499	25035.498
Observations	14,250	14,250	6,178	6,178

Notes: Column (3) restricts the sample to short-tenure workers (tenure $\leq$ 2), defined as workers with tenure 0–2 years in their current job (tenure = survey year − job start year). All statistics are computed using sampling weights. Standard deviations are reported in columns (2) and (4). Education level is 2 for absence of schooling, 3 for elementary school, 4 for middle school, 5 for high school, 6 for college, 7 for bachelor's degree, 8 for master's degree, and 9 for doctoral degree. Housing tenure type denotes the contract type of housing with 1 for owner, 2 for 'Jeonsei', 3 for monthly renter, and 4 for others. Housing structure type denotes the type of housing construction with 1 for detached house, 2 for apartment, 3 for townhouse, 4 for multi-unit house, 5 for flats with shops, and 6 for etc.

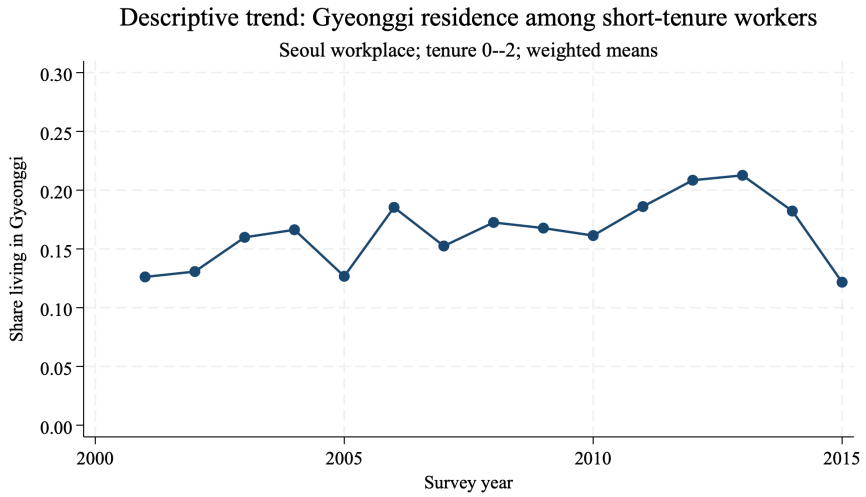


Figure 1: Gyeonggi residence among short-tenure workers over time

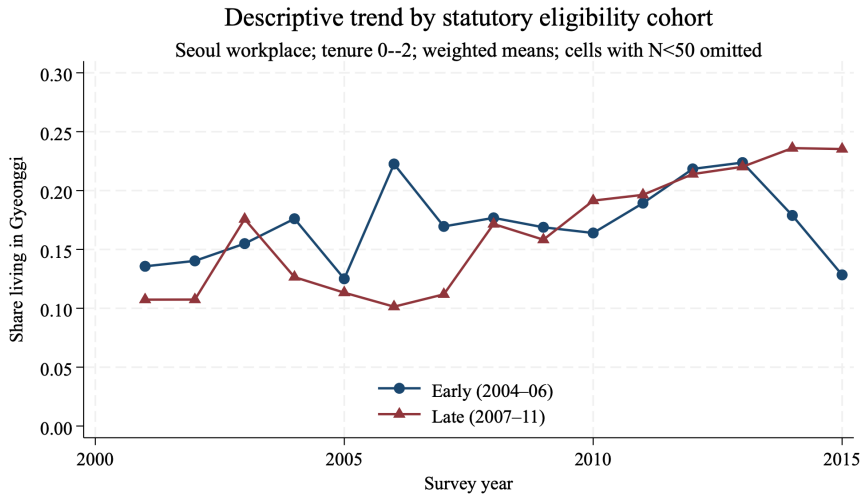


Figure 2: Gyeonggi residence by statutory eligibility cohort (Early vs Late)

Figure 1 plots the weighted share of short-tenure Seoul-workplace workers who reside in Gyeonggi by survey year. The series exhibits a gradual upward drift over the sample period, indicating a broad increase in Seoul workplace–Gyeonggi residence matches. At the same time, the raw series reflects not only policy timing but also changes in cohort composition, housing-market conditions, and broader labor-market trends, and should therefore be interpreted as descriptive rather than causal.

Figure 2 decomposes the same outcome by statutory eligibility cohort, comparing early-covered cohorts (2004–06) with late-covered cohorts (2007–11). Cohort–year cells with fewer than 50 observations are omitted to reduce noise from sparse cells. The cohort-specific raw series do not display parallel trends, which cautions against interpreting simple cross-cohort differences as policy effects. This motivates the regression analysis in the next section, which exploits entry-based statutory timing while absorbing rich time effects and group-specific sources of heterogeneity.

In the next section, I formalize these descriptive patterns using entry-based ITT specifications that exploit staggered statutory timing and incorporate rich fixed effects, flexible time controls, and clustered inference.

5 Individual-Level Evidence

The spatial equilibrium framework discussed in Section 3 emphasizes that reductions in commuting disutility operate primarily through changes in the effective elasticity of labor supply across locations. While this elasticity is ultimately an aggregate object, its underlying adjustment mechanisms can be studied at the individual level. In particular, individual data make it possible to identify which margins of adjustment become more responsive following the reform. These margins correspond closely to the behavioral channels emphasized in the spatial equilibrium framework of Section 3, although the individual-level estimates are not interpreted as direct estimates of the aggregate elasticity. This section therefore provides individual-level evidence on how spatial adjustment responds to the Five-Day Workweek reform.

As discussed in Section 4, short-tenure workers make residential and commuting decisions under substantially fewer legacy constraints. Precisely because they are closer to the "re-optimization" margin, short-tenure workers provide a cleaner window into how spatial labor supply adjusts when commuting costs are plausibly reduced.

A central identification challenge is that realized reductions in weekend work at the worker–firm level may be endogenous. Firms may adjust schedules earlier or later than required by the legal mandate, compliance may be imperfect, and the reform may coincide with other changes in hiring, personnel policy, or workplace organization.[6] In addition, short-tenure workers jointly sort across jobs and residences, so post-reform patterns may reflect both statutory timing and time-varying sorting. For these reasons, I do not interpret the individual-level estimates as

[6] I do not observe firm-level take-up of the five-day workweek, and realized changes in working schedules may be correlated with other firm adjustments.

treatment effects of realized schedule changes.

Instead, the empirical strategy relies on entry-based intention-to-treat (ITT) exposure defined by statutory eligibility. Specifically, I exploit the quasi-experimental variation generated by the legal rollout, under which workers are differentially assigned to statutory eligibility cohorts based on the size and sector of their entry firms. Under the maintained assumption that, conditional on flexible controls and time effects, the timing of statutory eligibility is orthogonal to unobserved determinants of residential choice, the resulting estimates are interpreted as reduced-form effects of entry-based legal eligibility timing on individual-level spatial outcomes.

Within this ITT framework, I proceed in three steps. First, I estimate whether legal eligibility increases the probability of a Seoul workplace–Gyeonggi residence match among short-tenure workers. Second, I examine heterogeneity across groups with different mobility constraints (e.g., marital status or wage levels) or commuting-residential trade-offs. Third, I assess whether the observed response is concentrated specifically in the Seoul–Gyeonggi residential margin rather than in non-Seoul residence more broadly.

5.1 Identification

The empirical strategy exploits the staggered statutory rollout of the Five-Day Workweek reform across establishment size and sector. For each worker, I assign an *entry-based statutory eligibility cohort* according to the legal schedule that applied to the size and sector of the worker's entry firm. This assignment yields a worker-fixed intention-to-treat (ITT) exposure measure, denoted by the statutory eligibility year.

An important feature of this design is that the assigned ITT cohort is based on *entry-firm* characteristics and is therefore fixed at the worker level, whereas a worker's *current* job spell may later differ from the entry spell. Accordingly, the treatment variable used in the baseline specifications should not be interpreted as contemporaneous treatment status at the current firm. Instead, it captures exposure to the reform through entry-based legal eligibility timing. The resulting estimand is therefore a reduced-form effect of *entry-based statutory eligibility timing*, rather than a treatment effect of realized schedule changes at the worker–firm level.

For each individual, I construct a *statutory eligibility year assigned at entry*, by mapping entry-firm characteristics into the statutory rollout schedule. In particular, workers are assigned to the earliest year in which their entry firm becomes legally covered based on (*i*) *entry firm-size category*, and (*ii*) *whether the entry job is in the public sector or finance*. This produces a set of fine eligibility cohorts aligned with the legal schedule.

Let E_i denote worker i's statutory eligibility year assigned at entry[7]. The ITT exposure indicator is then defined as

$$D_{it} = \mathbf{1}\{E_i > 0\}\mathbf{1}\{t \geq E_i\}. \tag{10}$$

[7] $E_i \in \{2004, 2005, 2006, 2007, 2008, 2011, 0\}$ is determined by the size and sector of the entry firm under the legal rollout schedule.

Thus, treatment varies with statutory timing rather than realized take-up.

The maintained identifying assumption is a conditional parallel-trends restriction at the level of statutory eligibility timing: conditional on industry-by-year fixed effects, firm-size-specific linear trends, and observed controls, the timing of legal coverage is orthogonal to unobserved determinants of the outcome. Under this assumption, differences in outcomes across cohorts around the onset of statutory eligibility can be interpreted as reduced-form causal effects of entry-based legal eligibility timing.

Within this framework, treatment and control status are defined by legal timing. Workers assigned to a positive statutory eligibility year are ITT-treated; workers assigned to no statutory eligibility within the legal schedule are never-treated. In the event-study specifications, identification comes from comparing workers who are already eligible at a given event time with workers who are not yet eligible or never eligible, after absorbing rich fixed effects and time controls. Thus, the comparison is not a simple treated-versus-untreated cross-section, but a comparison across workers facing different statutory eligibility timing within the same Seoul workplace market and within a short-tenure sample.

Several concerns may threaten the plausibility of this assumption, and the design addresses them as follows.

First, statutory eligibility is mechanically linked to firm size and sector, so treated and untreated workers may differ systematically in both observed and unobserved characteristics. Balance tests confirm that unconditional comparability is imperfect (see Table A1 in the Appendix), especially between ever-treated and never-treated workers. In the short-tenure Seoul workplace sample, ever-treated workers are about 2.5 years younger, approximately 0.6 education-category points more educated, and about 0.19 log points higher-wage than never-treated workers when characteristics are measured at the worker's entry observation when available, or otherwise at the first observed year. Differences are smaller when comparing earlier-treated cohorts (2004–06) with later-treated cohorts (2007–11), but are not eliminated. I therefore do not rely on unconditional balance across groups for identification.

Second, industry-level shocks may line up mechanically with treatment timing because legal coverage is partly determined by sector and firm size. I address this by including industry-by-year fixed effects, so that identification comes from within-industry differences in statutory timing rather than from cross-industry changes.

Third, early- and late-covered cohorts may follow different smooth trajectories even absent the reform. To reduce bias from such differential evolution, I allow flexible trend controls, including cohort-specific trends in robustness exercises. This shifts identification toward deviations around the onset of statutory eligibility rather than long-run level differences across cohorts.

Fourth, workers in the short-tenure sample may sort jointly across jobs and residences, and the extent of such sorting may vary over time. Focusing on short-tenure workers does not eliminate selection; rather, it reduces confounding from legacy constraints embedded in pre-existing job–housing matches and makes the relevant adjustment margin more transparent.

Defining exposure using entry-based legal eligibility further limits mechanical endogeneity from post-entry job changes and treatment reclassification. Moreover, evidence on mobility frictions and monopsony in Korea suggests that rapid anticipatory resorting into covered firms is less plausible than in highly fluid labor markets. (See Hong (2022)) This does not eliminate sorting, but it makes it less likely that the estimates are driven solely by abrupt compositional shifts at eligibility onset.

Finally, other contemporaneous changes (e.g., infrastructure expansion or large-scale housing development) could generate similar spatial patterns. During the 2000s, the Seoul metropolitan area experienced major housing-development initiatives that could have affected residential choices independently of the reform.[8] The metropolitan bus system was also reorganized in 2004 and expanded throughout the 2000s, plausibly reducing commuting frictions from Gyeonggi to Seoul. To mitigate these concerns, I complement the baseline specifications with cohort-based event studies and placebo/falsification exercises using pseudo-eligibility dates and legally non-exposed workplace markets. If the estimated patterns are driven primarily by common infrastructure shocks, similar timing patterns should appear in these placebo settings.

Taken together, these design choices imply that the individual-level estimates are interpreted as reduced-form evidence on how entry-based statutory eligibility timing affects spatial outcomes among short-tenure workers in the Seoul labor market. They do not identify the direct effect of realized reductions in working days at the current employer, but they help distinguish statutory timing effects from several leading alternative explanations.

5.2 Empirical Strategy

The baseline regression is specified as

$$Y_{it} = \beta D_{it} + X_{it}'\gamma + \alpha_i + \delta_{j(i,t),t} + \phi_{s(i,t)} \cdot t + \lambda_{o(i,t)} + \varepsilon_{it}, \tag{11}$$

where the sample is restricted to workers employed in Seoul in calendar year t, and the outcome Y_{it} equals one if individual i resides in Gyeonggi in year t and zero otherwise. In the Seoul-workplace sample, this outcome captures the incidence of Seoul workplace–Gyeonggi residence matches. The variable D_{it}^{ITT} is the entry-based ITT exposure indicator defined in equation (10). The coefficient β therefore captures the average post-eligibility difference in the probability of residing in Gyeonggi among legally exposed workers, relative to workers who are not yet eligible or never eligible, conditional on the included controls and fixed effects. As discussed above, this coefficient is interpreted as a reduced-form effect of entry-based statutory eligibility timing rather than as the contemporaneous effect of realized changes in working days at the current employer.

The vector X_{it} includes time-varying worker characteristics, such as age, age squared, edu-

[8] According to Korea Land and Housing Corporation materials on second-phase new-town development, the major Greater Seoul Area projects included Pangyo, Paju, Dongtan, Gimpo, Geomdan, Songpa, Pyeong-Taek, and Gwanggyo.

cation, sex, marital status, child status, and log wages. The term α_i denotes individual fixed effects. The term $\delta_{j(i,t),t}$ denotes industry-by-year fixed effects, which absorb shocks common to industry j in year t, including sectoral demand shifts, restructuring, and changes in workplace geography. The term $\phi_{s(i,t)} \cdot t$ captures firm-size-category-specific linear trends, implemented as interactions between current firm-size-category indicators and calendar year. Finally, $\lambda_{o(i,t)}$ denotes occupation fixed effects; because occupation can vary over time in the data, these effects are not absorbed by the individual fixed effects.

Since residential location choices may adjust with delay due to relocation costs, I also estimate dynamic effects using an event-study specification:

$$Y_{it} = \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k \mathbf{1}\{t - E_i = k\} + X_{it}'\gamma + \alpha_i + \delta_{j(i,t),t} + \phi_{s(i,t)} \cdot t + \lambda_{o(i,t)} + \varepsilon_{it}, \tag{12}$$

where E_i denotes the worker's statutory eligibility year assigned at entry, event time is defined as $t - E_i$, and the event window is binned to $\mathcal{K} = \{-4, -3, -2, 0, 1, 2, 3, 4\}$. The period $k = -1$ is omitted as the reference category. The coefficients $\{\beta_k\}$ trace the evolution of outcomes relative to the year immediately preceding statutory eligibility. In this framework, the absence of systematic pre-eligibility effects provides evidence consistent with the identifying assumption.

5.3 Short-Tenure Workers' Residential Adjustment

This subsection examines whether short-tenure workers in the Seoul labor market become more likely to reside in Gyeonggi after entry-based statutory eligibility for the Five-Day Workweek reform. Table 3 reports estimates from the baseline specification in equation (11) and the event-study specification in equation (12). All columns restrict the sample to wage workers employed in Seoul with tenure in $[0, 2]$, and include individual fixed effects, occupation fixed effects, time-varying worker controls, and firm-size-specific linear trends. Column (1) includes year and industry fixed effects, while Column (2) replaces these with industry-by-year fixed effects. Column (3) reports the corresponding event-study specification using the richer industry-by-year fixed-effects structure.

Across Columns (1) and (2), the ITT coefficient captures the average change in the probability of residing in Gyeonggi after statutory eligibility begins, relative to periods before eligibility or to never-eligible workers, conditional on the included controls and fixed effects. The estimates are positive and statistically significant. In the preferred specification with industry-by-year fixed effects in Column (2), legal eligibility is associated with a 5.2 percentage point higher probability of residing in Gyeonggi while working in Seoul. This implies a higher prevalence of Seoul workplace–Gyeonggi residence matches, consistent with the spatial mechanism in Section 3 whereby lower commuting intensity makes longer-distance commuting more attractive at the entry margin.

Table 3: Impact of the Five-Day Workweek on Gyeonggi Residence (Short-Tenure Workers)

	(1) ITT: Year + Ind FE	(2) ITT: Year × Ind FE	(3) Event study: Year × Ind FE
Post (ITT): $t \geq$ eligibility year	0.032*	0.052**	
	(0.016)	(0.024)	
Avg. lead (t=−4 to −2)			0.014
			(0.022)
Avg. lag (t=0 to +4)			0.065**
			(0.028)
Observations	5,380	5,380	5,380
R^2	0.895	0.943	0.943
Occupation FE	O	O	O
p-value: leads joint test			0.881
p-value: lags joint test			0.018

Notes: Standard errors in parentheses. Sample: wage workers employed in Seoul workplaces with tenure in $[0, 2]$. The ITT exposure year is assigned mechanically from entry firm size and sector under the legal rollout schedule. Event time is defined as calendar year minus the ITT eligibility year; the reference period $t = -1$ is omitted in the event-study regression. Standard errors are clustered at the individual level; regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (3) reports the event-study specification, which traces residential adjustment around the onset of statutory eligibility. The average post-eligibility effect, computed over event times $k = 0$ to $+4$, is positive and economically meaningful (0.065), and the joint test rejects the null of no post-eligibility effects ($p = 0.018$). By contrast, the average pre-eligibility effect over $k = -4$ to -2 is small and statistically indistinguishable from zero, and the joint test does not reject the null of no pre-eligibility effects ($p = 0.881$). Taken together, these patterns are consistent with the baseline ITT estimates and do not indicate clear evidence of differential pre-trends prior to statutory eligibility.

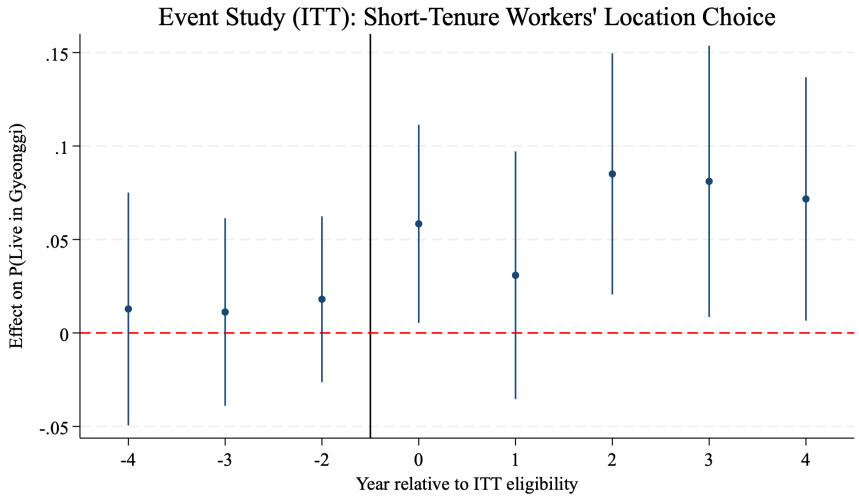


Figure 3: Event study dynamics of Gyeonggi residence among short-tenure workers

Figure 3 plots the event-time coefficients from equation (12). The lead coefficients are small

and imprecisely estimated, whereas the post-eligibility coefficients are uniformly positive and become larger after eligibility begins. This pattern is consistent with a residential response that emerges after legal coverage and persists for several years, rather than with a sharp pre-existing divergence across cohorts. The fact that the post-eligibility effects remain positive over several subsequent years is also consistent with gradual adjustment in residential location, plausibly reflecting relocation frictions rather than instantaneous re-optimization.

5.4 Robustness Checks

This subsection evaluates the sensitivity of the individual-level findings to alternative identifying concerns, estimators, and specification choices. The baseline results suggest that entry-based statutory eligibility for the Five-Day Workweek increased the likelihood that short-tenure workers employed in Seoul resided in Gyeonggi, and the event-study estimates do not show clear pre-eligibility deviations. However, these findings remain subject to several familiar concerns. In particular, the estimated pattern could still reflect spurious policy timing, broader metropolitan shocks that happen to coincide with the statutory rollout, or sensitivity to specific modeling choices in the event-study design.

To address these concerns, this subsection presents a set of complementary robustness checks. I first report placebo and validation tests designed to assess whether similar timing patterns appear in settings where the statutory rollout should have little or no explanatory power. I then examine the sensitivity of the main results to alternative sample restrictions and specification choices, including a narrower entrant sample, alternative event-study windows, and alternative trend controls. Finally, I report estimates based on an alternative staggered-adoption DiD procedure as an additional check on the dynamic pattern. Across these exercises, the main question is whether the post-eligibility response remains positive while pre-eligibility patterns remain limited.

Table 4: Placebo and Validation Tests

	(1) Fake reform 2002	(2) Fake reform 2003	(3) Small firms placebo	(4) Non-Seoul workplace: Post ITT	(5) Non-Seoul workplace: Event study
Placebo effect	0.002	-0.009			
	(0.009)	(0.012)			
Legal coverage (ITT)			-0.003	0.002	
			(0.010)	(0.012)	
Avg. lead (t=−4 to −2)					-0.021
					(0.016)
Avg. lag (t=0 to +4)					-0.007
					(0.016)
Observations	3,237	3,237	488	5,465	5,465
R^2	0.971	0.971	0.943	0.970	0.970
p-value: leads joint test					0.142
p-value: lags joint test					0.862

Notes: Standard errors in parentheses. Unless otherwise noted, the sample consists of wage workers employed in Seoul workplaces with tenure in $[0, 2]$. Columns (1)–(2) restrict the sample to pre-2004 years and assign fake reform years (2002 and 2003, respectively), treating the early legal cohorts as pseudo-treated. Column (3) restricts the sample to very small firms, defined as establishments with fewer than five workers, which are treated as legally non-covered. Columns (4)–(5) use a placebo sample of non-Seoul metropolitan workplaces (e.g., Gyeonggi and Incheon), with the outcome defined as residing in Seoul. Column (4) reports the post-ITT coefficient, while Column (5) reports the corresponding event-study summary. Event time is defined relative to the worker-fixed ITT eligibility year assigned from entry firm size and sector; the reference period is $t = -1$. Standard errors are clustered at the individual level, and regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports placebo and validation tests designed to assess whether the baseline estimates are driven by spurious timing patterns or by broader metropolitan shocks unrelated to statutory exposure.

Columns (1) and (2) implement pre-reform placebo exercises by assigning fake reform years (2002 and 2003, respectively) and treating the early legal cohorts as pseudo-treated. In both cases, the estimated placebo effects are small and statistically indistinguishable from zero. These null results reduce the concern that the baseline findings are generated by artificial timing breaks that would have appeared even before the actual statutory rollout.

Column (3) restricts the sample to very small firms, which are plausibly not covered under the legal schedule. In this legally non-covered sample, the estimated post-eligibility effect is again close to zero and statistically insignificant. Although this placebo does not by itself rule out all metropolitan-wide shocks, since very small firms remain exposed to the same urban environment, it weakens the interpretation that the baseline estimates are driven by common time shocks unrelated to statutory exposure.

Columns (4) and (5) report exercises using workers employed in non-Seoul metropolitan workplaces, with the outcome defined as residing in Seoul. Column (4) shows that the post-ITT coefficient is small and insignificant, while Column (5) shows no systematic event-time pattern around statutory eligibility: the average lead and lag effects are both close to zero, and the joint

Table 5: Robustness to Alternative Samples and Specifications

	(1) Baseline event study	(2) New entrants: Post ITT	(3) Event study window: $[-3, +3]$	(4) Cohort-specific trends	(5) Entry-residence × year FE
Legal coverage (ITT)		0.071**			
		(0.032)			
Avg. lead	0.014		0.014	0.022	0.021
	(0.022)		(0.021)	(0.021)	(0.022)
Avg. lag	0.065**		0.063**	0.049*	0.064**
	(0.028)		(0.028)	(0.032)	(0.027)
Observations	5,380	1,937	5,380	5,380	5,380
R^2	0.943	0.222	0.943	0.943	0.945
p-value: leads joint test	0.881		0.729	0.687	0.634
p-value: lags joint test	0.018		0.015	0.064	0.021

Notes: Standard errors in parentheses. Unless otherwise noted, the sample consists of wage workers employed in Seoul workplaces with tenure in $[0, 2]$. Column (1) reports the baseline event-study specification from the main text. Column (2) restricts the sample to new entrants (tenure = 0) and estimates the post-ITT coefficient using year and industry fixed effects. Column (3) re-estimates the event study using a shorter event window, $[-3, +3]$. Column (4) replaces firm-size-specific linear trends with eligibility-cohort-specific linear trends. Column (5) augments the baseline specification with entry-residence-group-by-year fixed effects to absorb residence-group-specific time shocks. For event-study specifications, "Avg. lead" is computed over the available pre-period coefficients and "Avg. lag" over the available post-period coefficients, with $t = -1$ omitted as the reference period. Standard errors are clustered at the individual level, and regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tests fail to reject the null of no pre- or post-eligibility effects. The full dynamic path for this placebo event study is reported in Appendix Figure 1. Taken together, these results make it less likely that the main findings are driven solely by metropolitan-wide shocks, such as common housing-market or transportation changes, rather than by the statutory timing itself.

Table 5 reports robustness checks based on alternative samples and specification choices, using the baseline event-study result in Column (1) as a benchmark.

Column (2) restricts the sample to new entrants (tenure = 0) and estimates a parsimonious post-ITT specification with year and industry fixed effects. Industry-by-year fixed effects are not included in this specification because entrant counts are sparse within many industry–year cells, leaving limited within-cell variation for identification. The positive and statistically significant ITT coefficient indicates that the post-eligibility increase in Gyeonggi residence is already visible at the immediate entry margin, and is not driven solely by the inclusion of workers with slightly longer tenure in the broader short-tenure sample.

Columns (3) and (4) examine the sensitivity of the event-study results to alternative modeling choices. Column (3) shortens the event window to $[-3, +3]$ instead of $[-4, +4]$. The average post-eligibility effect remains positive and statistically significant, while the average lead remains small and statistically indistinguishable from zero. Column (4) replaces firm-size-specific linear trends with eligibility-cohort-specific linear trends. Under this alternative trend specification, the average post-eligibility effect remains positive, although somewhat attenuated and

only marginally significant, while the average lead remains close to zero. These results suggest that the baseline pattern is not driven by a particular choice of event window or trend control.

Column (5) augments the baseline event-study specification with entry-residence-group-by-year fixed effects. This specification absorbs time-varying shocks common to workers with different initial residence locations, such as differential trends across entry-Seoul, entry-Gyeonggi, and entry-other groups. The average post-eligibility effect remains positive and statistically significant, while the average lead remains small and insignificant. The corresponding event-time coefficients are shown in Appendix Figure 2. This indicates that the baseline result is not explained simply by different time trends across workers with different entry residence locations.

Taken together, the estimates in Table 5 show that the main individual-level finding is robust to alternative sample restrictions, event-study windows, trend specifications, and additional controls for entry-residence-specific time effects.

As an additional check, I estimate dynamic effects using the Callaway and Sant'Anna (2021) procedure. Because the empirical design is based on worker-fixed entry eligibility rather than a canonical staggered-adoption treatment defined at the current employer, this exercise should be viewed as supplementary.[9] The estimated coefficients are imprecise, although the post-treatment effects are mostly positive. I therefore treat these results as broadly consistent with, but much weaker than, the main event-study evidence. The corresponding figure is reported in Appendix Figure 3.

6 Heterogeneity and Mechanisms

The baseline estimates in Section 5 identify an average increase in Seoul workplace–Gyeonggi residence matches around statutory eligibility. This section asks two follow-up questions. First, is the response stronger for some workers than for others? Second, does the increase in Gyeonggi residence reflect a shift away from Seoul residence or a broader expansion of non-Seoul residence? Section 6.1 addresses the first question through heterogeneity analyses, and Section 6.2 addresses the second through a decomposition of residence outcomes.

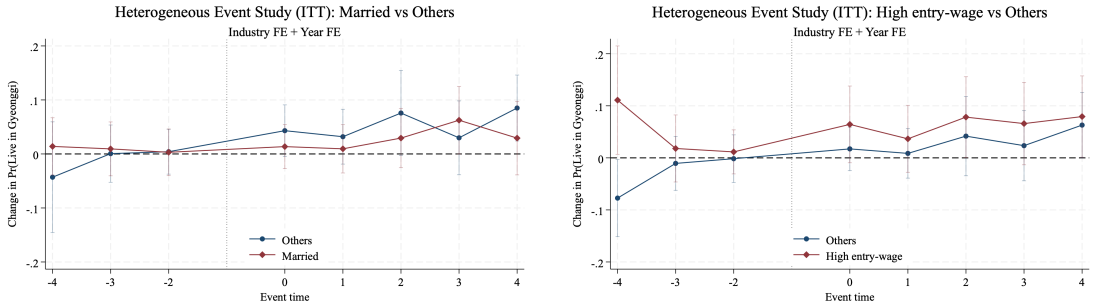
6.1 Heterogeneity

The baseline results in Section 5.3 show that short-tenure workers employed in Seoul become more likely to reside in Gyeonggi around statutory eligibility. To assess heterogeneity, I interact the ITT event-time indicators with subgroup indicators for marital status, entry-wage

[9] The treatment variable used in the main analysis is an intention-to-treat measure based on the statutory eligibility year assigned from entry-firm characteristics. Hence treatment timing is fixed at the worker level and may differ from contemporaneous realized exposure at the current employer. In particular, if workers move across firms after entry, the assigned eligibility date continues to reflect entry conditions rather than the timing at which the current employer becomes covered. For this reason, the mapping from treatment timing to observed exposure does not coincide exactly with the canonical staggered-adoption framework assumed in standard multi-period DiD estimators.

group, and residence location at current-job entry, while keeping the same sample restrictions, controls, and weighting scheme as in Section 5.3.

A practical difference is the fixed-effects specification. For heterogeneity by marital status and entry wage, I use separate industry and year fixed effects, together with individual and occupation fixed effects and firm-size-specific linear time trends. This choice avoids rank-deficiency problems that arise in saturated industry-by-year specifications when subgroup-by-event-time variation is sparse within some industry-year cells, especially for the entry-wage split. By contrast, the heterogeneity analysis by residence location at current-job entry is estimated with the stricter industry-by-year fixed-effects specification. Accordingly, the heterogeneity results should be interpreted as descriptive evidence, with somewhat stronger time-variation controls in the entry-residence analysis than in the marital-status and entry-wage analyses.



(a) Heterogeneity by marriage status

(b) Heterogeneity by wage group

Figure 4: Heterogeneity among short-tenure workers

Figure 4a shows heterogeneous event-study patterns by marital status. For both groups, pre-coverage coefficients are close to zero and relatively imprecisely estimated, whereas post-coverage coefficients become positive. The post-coverage increase appears more gradual for married workers, while non-married workers exhibit a sharper rise shortly after coverage. This pattern is suggestive of higher household-level moving costs and coordination frictions among married households, although the estimates are best interpreted as descriptive evidence on adjustment dynamics rather than as precise structural parameters.

Figure 4b presents heterogeneity by entry wage, where the high-wage group is defined as the top tercile of the entry-wage distribution. The high-wage group exhibits larger point estimates than the non-high-wage group in both the event-study plot and the post-coverage averages, although the difference in average post-coverage responses is not precisely estimated. A suggestive interpretation is that reductions in commuting disutility disproportionately relax effective commuting constraints for workers in higher-paying entry jobs—potentially because these jobs are more concentrated in Seoul and involve longer baseline commutes—making suburban residence relatively more attractive at entry. An alternative interpretation is compositional: high-wage entry jobs may be concentrated in specific industries or occupations with distinct geographic recruitment patterns; the inclusion of industry and year fixed effects and

Table 6: Heterogeneity in Pre/Post Event-Study Averages (Short-Tenure Workers)

	Panel A: Marital status		Panel B: Entry wage	
	Others	Married	Others	High entry-wage
Avg. lead (t=-4 to -2)	-0.013	0.009	-0.030	0.047
	(0.027)	(0.021)	(0.025)	(0.028)
Avg. lag (t=0 to +4)	0.053**	0.029	0.031*	0.065
	(0.023)	(0.022)	(0.021)	(0.033)
Diff. in avg lag		-0.024		0.034
SE		0.024		0.033
p-value: diff.		0.318		0.299
Observations	5,701	5,701	5,266	5,266
R^2	0.895	0.895	0.883	0.883

Notes: Sample: wage workers employed in Seoul workplaces with tenure in $[0, 2]$. All specifications include industry and year fixed effects, firm-size-specific linear trends (firm size × year), and individual and occupation fixed effects. "Avg. lead" is computed as the mean of event-time coefficients for k=-4,-3,-2; avg. lag for k=0,...,4. Base period k=-1 omitted. "Diff. in avg lag" is the difference in average post-coverage effects across groups (Groups − Others); the standard error and p-value are for the null of no difference. Standard errors are clustered at the individual level; regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

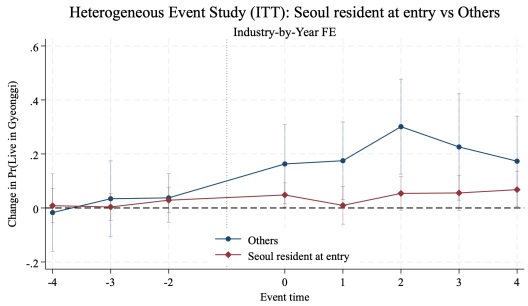
occupation fixed effects mitigates, but may not fully eliminate, such differences.

Table 6 summarizes the event-study paths using average leads ($k = -4, -3, -2$) and average lags ($k = 0, \dots, 4$), and reports the difference in average lags across groups.

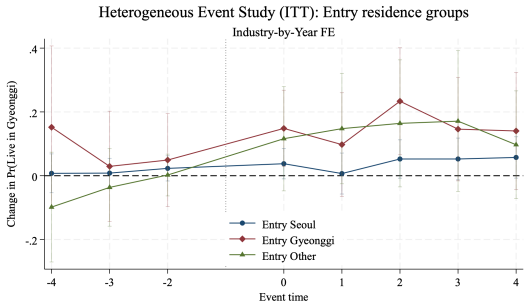
By marital status, average leads are close to zero for both groups (Others: -0.013 with SE 0.027; Married: 0.009 with SE 0.021). Average lags are positive for both groups, and statistically significant for non-married short-tenure workers (0.053 with SE 0.023), while the married estimate is smaller and not statistically significant (0.029 with SE 0.022). The post-period difference (Married minus Others) is -0.024 (SE 0.024; $p = 0.318$), indicating limited evidence that marital status differentially shifts the post-coverage response on average.

By entry wage, the high-wage group shows larger positive post-coverage point estimates than the non-high-wage group. Average lags are 0.031 (SE 0.021) for others and 0.065 (SE 0.033) for the high-wage group, but the difference in average lags is not precisely estimated (0.034 with SE 0.033; $p = 0.299$). On the pre-coverage side, the high-wage group's average lead is positive but not statistically distinguishable from zero. Overall, the point estimates are suggestive of stronger adjustment among high-wage workers, but this difference is not precisely estimated in the current specification and should be interpreted cautiously.

I also examine heterogeneity by residence location at current-job entry. In contrast to the marital-status and entry-wage analyses, these specifications are estimated with industry-by-year fixed effects. Figure 5a compares workers who resided in Seoul at entry with all others, and shows that the post-eligibility increase in Gyeonggi residence is larger for the non-Seoul-at-entry group. Appendix Table A2 summarizes the same pattern numerically: the average post-eligibility effect is 0.208 (SE 0.078) for the non-Seoul-at-entry group, compared with 0.047 (SE 0.026) for the Seoul-at-entry group, and the difference in average post-eligibility effects is -0.161 (SE 0.074; $p = 0.031$). Figure 5b further decomposes the sample into three entry-residence



(a) Heterogeneity by Seoul residence at current-job entry



(b) Heterogeneity by entry residence group

Figure 5: Heterogeneity by residence location at current-job entry

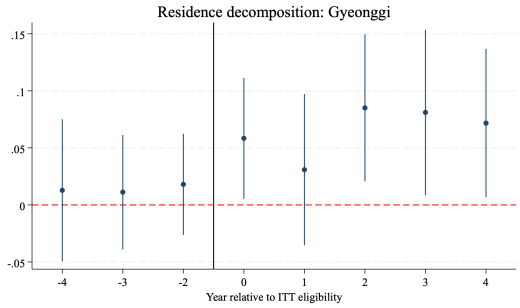
groups—Seoul, Gyeonggi, and other regions—and shows that the larger post-eligibility response is concentrated among workers whose entry residence was already in Gyeonggi or outside Seoul, whereas the Seoul-at-entry group exhibits relatively modest changes.

These patterns suggest that the main individual-level effect is not driven primarily by workers exiting Seoul residence after eligibility. Rather, the increase in Seoul workplace–Gyeonggi residence matches appears stronger among workers who were already located outside Seoul at the beginning of the current job match. One interpretation is that reduced commuting intensity makes it easier for such workers to retain or choose non-Seoul residence while working in Seoul. At the same time, entry residence is likely to proxy for broader differences in household constraints, housing-market attachment, and pre-existing geographic ties. These results should therefore be interpreted as descriptive heterogeneity rather than as definitive evidence on the exact migration mechanism.

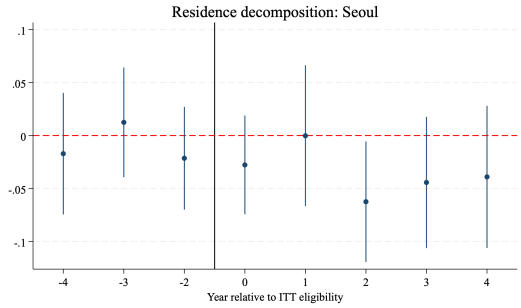
6.2 Residence Decomposition

The heterogeneity results in Section 6.1 indicate that the response to statutory eligibility is not uniform across workers. A related question is what this response represents in residential terms. In particular, the baseline increase in Seoul workplace–Gyeonggi residence matches could reflect a shift away from Seoul residence toward Gyeonggi, a broader increase in non-Seoul residence that happens to be concentrated in Gyeonggi, or some combination of the two. To distinguish among these possibilities, this subsection decomposes the residence response by residence category.

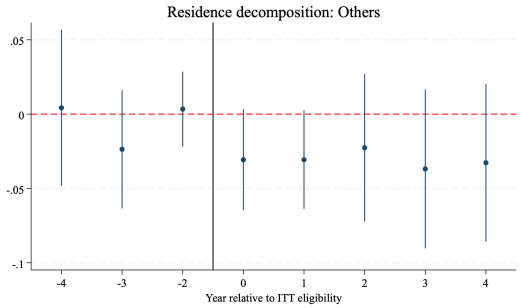
Figure 6 decomposes the residence response by residence location, showing event-study patterns for workers who live in Gyeonggi (Panel A), Seoul (Panel B), and other regions (Panel C). The Gyeonggi-residence equation in Panel A uses the same outcome as the baseline specification in Section 5.3 and therefore reproduces the main result in a decomposed setting. Appendix Table A3 reports the corresponding numerical summaries. Following statutory eligibility, the probability of residing in Gyeonggi rises: the post-only coefficient is 0.052 (SE 0.024), and the average post-eligibility effect is 0.065 (SE 0.028). By contrast, the probability of residing



(a) Workers who live in Gyeonggi



(b) Workers who live in Seoul



(c) Workers who live outside Seoul and Gyeonggi

Figure 6: Decomposition of residence responses by residence location

in Seoul declines, with a post-only coefficient of -0.027 (SE 0.022) and an average post-eligibility effect of -0.035 (SE 0.025); the joint test for the post-eligibility coefficients rejects the null of no response at the 5 percent level. Residence in other regions also shows negative point estimates, but they are smaller and not jointly significant: the post-only coefficient is -0.024 (SE 0.015), the average post-eligibility effect is -0.031 (SE 0.021), and the joint post-period test yields a p-value of 0.432.

Taken together, these patterns indicate that the main individual-level effect is concentrated on the Seoul–Gyeonggi residential margin. The decomposition suggests that the baseline increase in Seoul workplace–Gyeonggi residence matches is accounted for mainly by a lower post-eligibility probability of residing in Seoul, rather than by a broad increase in residence outside both Seoul and Gyeonggi. Importantly, this does not imply that the adjustment is driven primarily by Seoul-at-entry workers moving out of Seoul. As shown in the entry-residence heterogeneity analysis, the increase in Gyeonggi residence is in fact larger among workers who were already located outside Seoul at current-job entry. Taken together, the two sets of results suggest that the overall response is concentrated on the Seoul–Gyeonggi margin, while its incidence is stronger among workers whose initial residential position was already outside Seoul. This interpretation is consistent with reduced commuting intensity making Gyeonggi residence relatively more attractive within the Seoul labor market. At the same time, the decomposition should not be interpreted as identifying exact worker-level migration paths; it is more appropriately read as evidence on which residence margin accounts for the main reduced-

form response.

7 Conclusion

This paper studies whether working-time regulation affects residential location choices within a metropolitan labor market. Using the phased statutory introduction of the Five-Day Workweek in South Korea, we assign each worker an eligibility year based on the size and sector of the worker's entry firm and examine residential adjustment around statutory coverage.

We find that short-tenure workers employed in Seoul become about 5 percentage points more likely to reside in Gyeonggi after statutory eligibility begins. The event-study estimates show little systematic evidence of differential pre-eligibility trends, while the post-eligibility coefficients are positive and jointly significant. The response is concentrated on the Seoul–Gyeonggi residential margin. The heterogeneity analysis further suggests that adjustment is not uniform across workers, with the clearest differences appearing by residence at current-job entry. Taken together, these findings are consistent with working-time regulation affecting residential location through changes in commuting-related incentives.

This paper contributes to the literature by adding a spatial dimension to the study of working-time regulation. While previous studies of the Korean Five-Day Workweek have focused primarily on health, employment, productivity, and other labor-market outcomes, this paper provides evidence that statutory changes in working time can also affect residential location within a metropolitan labor market. More broadly, the results suggest that labor-market institutions may have spatial consequences that extend beyond their conventional effects on hours, employment, and worker well-being.

These findings are also relevant to ongoing discussions of further reductions in working time. The results suggest that evaluations of shorter-workweek policies may need to consider residential location and other spatial adjustment margins in addition to conventional labor-market outcomes. Future work could examine longer-run adjustments in residential and workplace location, housing markets, and commuting behavior, as well as directly measure how changes in work schedules translate into changes in commuting patterns.

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Appendix

A1. Additional Tables and Figures

Table A1: Baseline Comparability across ITT Groups

Variable	Mean G0	SD G0	Mean G1	SD G1	Diff.	Std. diff.	*p*	*N*
Panel A. Treated vs. Untreated								
Age	37.770	13.079	35.270	12.938	-2.500	-0.192	0.005	244 / 2118
Male	0.451	0.499	0.522	0.500	0.071	0.142	0.035	244 / 2118
Education	5.221	1.214	5.814	1.320	0.593	0.468	0.000	244 / 2118
Married	0.463	0.500	0.407	0.491	-0.057	-0.114	0.093	244 / 2118
Child	0.000	0.000	0.006	0.078	0.006	0.111	0.000	244 / 2118
Log wage	4.553	0.606	4.739	0.669	0.186	0.292	0.000	240 / 2108
Housing tenure type	1.674	0.847	1.583	0.808	-0.091	-0.110	0.113	239 / 2082
Housing structure type	2.569	1.323	2.314	1.240	-0.255	-0.199	0.004	239 / 2082
Floor space	27.677	16.957	30.339	19.545	2.662	0.146	0.150	96 / 872
Panel B. Early- vs. Late-Treated Cohorts								
Age	35.686	13.096	35.595	11.466	-0.091	-0.007	0.871	2111 / 555
Male	0.498	0.500	0.544	0.498	0.046	0.093	0.052	2111 / 555
Education	5.868	1.327	5.557	1.202	-0.312	-0.246	0.000	2111 / 555
Married	0.392	0.488	0.477	0.500	0.086	0.173	0.000	2111 / 555
Child	0.006	0.075	0.009	0.095	0.003	0.039	0.443	2111 / 555
Log wage	4.763	0.687	4.803	0.532	0.040	0.065	0.141	2104 / 552
Housing tenure type	1.604	0.813	1.639	0.856	0.035	0.042	0.388	2076 / 540
Housing structure type	2.325	1.226	2.365	1.254	0.040	0.032	0.505	2076 / 540
Floor space	30.638	20.380	28.509	15.309	-2.129	-0.118	0.087	867 / 220

Notes: One observation per individual is used. The entry-year observation is retained when available; otherwise the first observed year is used. In Panel A, Group 0 denotes never-treated workers and Group 1 denotes workers with a positive entry-based statutory eligibility year. In Panel B, Group 0 denotes early-treated cohorts (2004–2006) and Group 1 denotes late-treated cohorts (2007, 2008, and 2011). Standardized differences are computed as mean differences divided by pooled standard deviations. The final column reports the non-missing sample sizes in Group 0 and Group 1, respectively.

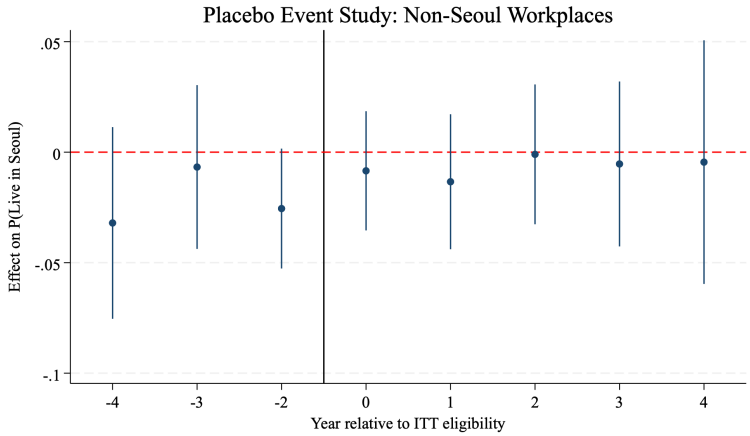


Figure A1: Placebo Event Study Estimates for Non-Seoul Workplace Sample

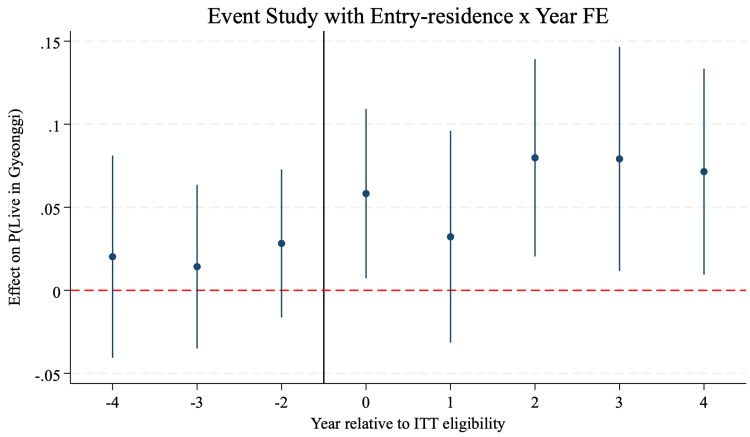


Figure A2: Event Study Estimates for Entry-Residence Group By Year Fixed Effect

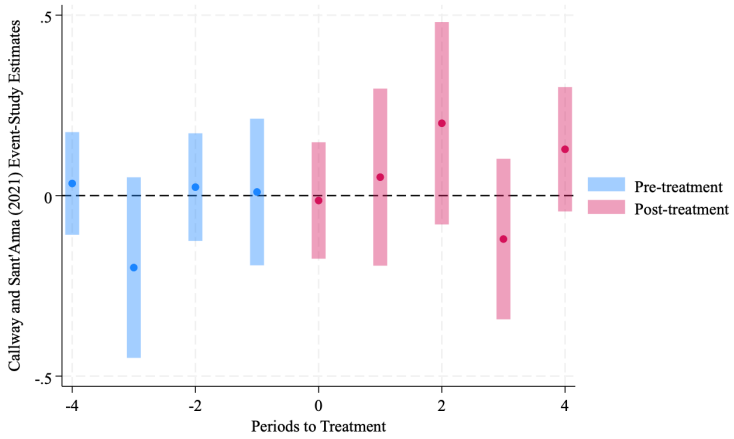


Figure A3: Callaway and Sant'Anna (2021) Event Study Estimates

Table A2: Heterogeneity by Seoul Residence at Current-Job Entry

	Others	Seoul resident at entry
Avg. lead ($t = -4$ to -2)	0.018	0.014
	(0.054)	(0.023)
p-value: leads	0.740	0.543
Avg. lag ($t = 0$ to $+4$)	0.208***	0.047*
	(0.078)	(0.026)
p-value: lags	0.008	0.035
Diff. in avg. lag (Seoul-at-entry − Others)	-0.161**	
SE	0.074	
p-value: diff.	0.031	
Observations	5,278	
R^2	0.937	

Notes: Sample: wage workers employed in Seoul workplaces with tenure in $[0, 2]$. The group indicator equals one for workers residing in Seoul at current-job entry, and zero otherwise. The specification includes industry-by-year fixed effects, firm-size-specific linear trends, and individual and occupation fixed effects. "Avg. lead" is computed as the mean of event-time coefficients for $k = -4, -3, -2$, and "Avg. lag" is computed for $k = 0, \ldots, 4$, with $k = -1$ omitted as the reference period. The rows labeled p-value: leads and p-value: lags report joint tests that all pre-eligibility and post-eligibility coefficients, respectively, are equal to zero; they are not the p-values associated with the average lead and average lag estimates. Standard errors are clustered at the individual level and regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Residence Decomposition of the Individual-Level Response

	(1) Gyeonggi	(2) Seoul	(3) Others
Panel A. Post (ITT)			
Post (ITT): $t \geq$ eligibility year	0.052**	-0.027	-0.024
	(0.024)	(0.022)	(0.015)
Panel B. Event-Study Specification			
Avg. lead ($t = -4$ to -2)	0.014	-0.009	-0.005
	(0.022)	(0.021)	(0.015)
p-value: leads	0.881	0.415	0.363
Avg. lag ($t = 0$ to $+4$)	0.065**	-0.035	-0.031
	(0.028)	(0.025)	(0.021)
p-value: lags	0.018	0.043	0.432
Observations	5,380	5,380	5,380
R^2	0.943	0.945	0.936

Notes: Standard errors in parentheses. Sample: wage workers employed in Seoul workplaces with tenure in $[0, 2]$. Each column reports the same specification using a different residence indicator as the dependent variable: residing in Gyeonggi, Seoul, or other regions. The post-only specification includes the post-ITT indicator, and the event-study summary reports average pre-eligibility coefficients over $t = -4, -3, -2$ and average post-eligibility coefficients over $t = 0, \ldots, 4$, with $t = -1$ omitted as the reference period. All specifications are estimated within the Seoul-workplace sample and R^2 is reported for the event-study specification in Panel B. The rows labeled p-value: leads and p-value: lags report joint tests that all pre-eligibility and post-eligibility coefficients, respectively, are equal to zero; they are not the p-values associated with the average lead and average lag estimates. Standard errors are clustered at the individual level, and regressions are weighted by the individual-constant mean weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.